Statement of interest – Xanadu Residency Program

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Area and topic. Research: *characterising the differences between quantum and classical machine learning models.* I would most like to work with the quantum machine learning team, ideally with Joseph Bowles, whose paper with Shahnawaz Ahmed and Maria Schuld, *Better than classical? The subtle art of benchmarking quantum machine learning models*, is the direct starting point of the project described below. I would also be glad to contribute PennyLane functionality or demos that come out of this work.

What I have done. To learn how to ask "where might a quantum model help?" carefully, I built a fair regression benchmark on four real datasets chosen along one axis: how periodic the inputs are. These are tidal water levels at San Francisco (inputs are astronomical phases), wind-turbine power (wind speed and direction), expected goals in soccer (shot geometry), and galaxy redshift from photometry (no periodic inputs, used as a control). I compared a data re-uploading circuit and a fidelity quantum kernel, both in PennyLane, with five classical models and with the methods each field actually uses: UTide harmonic analysis, IEC-style power curves, logistic xG and kNN photo-z. Every model had the same Optuna budget, cross-validation respected time blocks and matches, and periodic inputs were encoded equivalently for both families: 850 runs in total, with 50 to 1000 training points.

What I found. No quantum model beat the best classical or domain baseline anywhere. The more interesting results are about *why*:

- **The feature map has to match the data.** The quantum kernel with an angle map is competitive on tides (test R^2 0.927 vs. 0.931 for a Gaussian process). The IQP map failed on periodic inputs: its $ZZ(x_j x_k)$ terms are not 2π-periodic. The tuner rejected IQP in every tides and wind run, but often chose it for the non-periodic photo-z data.
- **Expressivity vs. trainability.** The variational circuit stalled on tides. Using the Fourier view of re-uploading models (Schuld, Sweke & Meyer), I computed least-squares "spectrum ceilings" and exact model spectra by FFT. A one-layer circuit's frequencies already allow $R^2 = 0.932$, so the gap was not missing frequencies. It came from training (a longer budget raised the circuit's mean R^2 from 0.870 to 0.894) and from the limited set of Fourier coefficients the ansatz can reach. Deeper circuits put about 30 % of their output at frequencies the data does not contain, and trained worse.
- My first hypothesis (that tides need higher harmonics) was wrong; the analysis showed it. I value that kind of result as much as a positive one.

What I would like to do at Xanadu. I would like to turn this kind of analysis into reusable tools and sharper questions. Examples: spectrum-ceiling diagnostics that tell a user in advance whether a circuit's frequencies suit their data; feature maps designed to respect periodic and mixed-type inputs; and the step from noiseless simulation to shot noise and hardware-realistic budgets. I am equally happy to work on the team's current questions; I care more about getting the comparison right than about the answer favouring quantum.

Why me. My coursework at Aalto covers quantum mechanics, quantum information, practical quantum computing and deep learning. As a research assistant intern at the Quantum AI Lab in Hanoi (2025) I worked with the quantum optimal control and superconducting-qubit teams. In a second project I raised a 6-qubit variational phishing classifier's accuracy from 77% to 89% by tracing its shortfall to the SPSA optimizer rather than the circuit, and showed with density-matrix simulation that depolarizing noise shifts its decision threshold while leaving its ranking (AUROC) almost unchanged. The habit behind both projects, finding out *why* a quantum model does or does not work, is what I would bring to the team. I have been a teaching assistant for Aalto courses in algorithms and in statistical physics, and at the Vietnam Quantum Summer School. I write tested, reproducible Python: the benchmark's code, 50 unit tests, and a paper whose numbers are all generated from the results are at [repository link].